MIL-T-36347(DM)
9 December 1964

MILITARY SPECIFICATION

TEST BRIDGE

1. SCOPE

1.1 This specification covers one type of test bridge and conductivity cell, suitable for the measurement of solution conductivity from 500 to 7000 micromhos per centimeter.

2. APPLICABLE DOCUMENTS

2.1 The following document, of the issue in effect on date of invitation for bids or request for proposal, forms a part of this specification to the extent specified herein:

STANDARD

Military

MIL-STD-105 Sampling Procedures and Tables for Inspection by Attributes.

(Copies of standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material. Material shall be of the highest quality throughout, and suitable in all respects for the purpose intended.

FSC 6630

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3.2 Design. The test bridge shall be a compact and rugged instrument suitable for operating on 115-volt, 60-cycle, alternating current. The measuring circuit shall be a Wheatstone bridge and shall have a vacuum tube null indicator. The bridge shall have a manual temperature compensator adjustable from 64° to 190° Fahrenheit (F.).

3.2.1 Cell component. The bridge shall be supplied with a dip-type conductivity cell constructed of high-temperature polystyrene with platinized gold-plated nickel rod electrodes. The cell shall have a constant of 2.

3.3 Performance. The test bridge shall read from 500 to 7000 micromhos per centimeter specific conductance referred to 25° centigrade, and shall have a midpoint of 1500 micromhos per centimer at which optimum sensitivity is found, when tested as specified in 4.4.1.

3.4 Accuracy. Accuracy of the conductivity reading shall be ± 2 percent, when tested as specified in 4.4.1.

3.5 Case. The bridge shall be housed in a sheet metal case, finished in grey enamel.

3.6 Identification marking. Each test bridge shall be legibly and permanently marked with the name or registered trademark of the manufacturer.

3.7 Service data. The contractor shall enclose, with each unit, two copies each of one or more booklets which contain complete instructions for installation, operation, maintenance, and a list of component parts. Wiring diagrams shall be included for electrical items or components. Each part of the unit other than common fasteners, etc., shall be identified in the service data and parts list by the manufacturer's part number. In addition, the contractor shall mail 15 copies of the service data to the contracting officer for distribution.

3.8 Workmanship. Workmanship shall be first class throughout. Test bridge shall be free from defects which detract from its appearance or may impair its serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual, auditory, etc., investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.2 Certificates of quality. When available, certificates of quality, supplied by the manufacturer of the component or material, listing the specified test method and test results obtained, may be furnished in lieu of actual performance of such testing by the contractor.

4.2 Sampling.

4.2.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105.

	Inspection level	AQL (percent defective)
For visual examination		
Major defects	II	1.0
Minor defects	II	2.5

4.2.2 For tests. Each test bridge shall be tested as specified herein.

Characteristic	Requirement	Test procedure	AQL
Performance	3.3	4.4.1	*
Accuracy	3.4	4.4.1	*

* Acceptance number shall be zero.

4.3 Examination. The test bridge shall be examined to determine compliance with all requirements contained in this specification.

4.3.1 Classification of defects. Examination shall be conducted in accordance with the following classification of defects:

Categories and defects*

Major

101 Component part not present.
102 Service data not present.
103 Service data not complete, correct, or legible.
104 Marking on dials not complete, correct, or legible.
105 Dials not clear.
106 Dials not free of cracks.
107 Calibration not as specified.
108 Case not free of sharp edges.
109 Electrodes not free of cracks.

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Minor

201 Case surface not smooth; not free of gouges, cracks, and scratches.
202 Dials not free of scratches.
203 Service data not free of tears and wrinkles.
204 Identification marking not present, complete, correct, legible, or permanent.

* Examination shall not be restricted to the classified possible defects listed above.

4.4 Tests. Tests shall be conducted to determine compliance with specification requirements.

4.4.1 Performance. The calibration of the test bridge shall be checked against a standard calibrated instrument certified by the National Bureau of Standards. Readings shall be checked at 500, 1500, and 7000 micromohos/centimeter in suitable test solutions at 64° F. and 190° F.

5. PREPARATION FOR DELIVERY

5.1 Packaging and packing. Level C. The commodity shall be packaged and packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.2 Marking.

5.2.1 Interior packages. Each package shall be marked to identify contents.

5.2.2 Exterior containers. Exterior containers shall be marked with stock number, name of material, quantity, name of consignee, name of consignor, and contract or purchase order number.

6. NOTES

6.1 Ordering data. Procurement documents should specify the number and date of this specification.

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6.2 This specification does not include all types, classes, grades, sizes, etc., of the commodity indicated by the title of this specification, or which are commercially available, but is intended to cover the types, etc., which are normally procured to meet Military requirements.

Custodians:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
Defense Supply Agency - DM

Review activities:
Army - MD
Navy - MS
Air Force - 03

Review information is current as of the date of this document. For future coordination of changes to this document, draft circulation should be based on the information in the current DODISS.

Category: C

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SPECIFICATION ANALYSIS SHEET

FORM APPROVED BUDGET BUREAU NO. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use of the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION MILITARY SPECIFICATION
TEST BRIDGE MIL-T-36347(EM), dated, 9 December 1964.

ORGANIZATION | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A ☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES" IN WHAT WAY?

4. REMARKS (ATTACH ANY PERTINENT DATA WHICH MAY BE OF USE IN IMPROVING THIS SPECIFICATION. IF THERE ARE ADDITIONAL PAPERS, ATTACH TO FORM AND PLACE BOTH IN AN ENVELOPE ADDRESSED TO PREPARING ACTIVITY)

SUBMITTED BY (PRINTED OR TYPED NAME AND ACTIVITY) | DATE

DD FORM 1426

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